

Contents

- [John Garcia, Andres Arcilia, Xavier Dammond](#)

%%Senior Project 1

John Garcia, Andres Arcilia, Xavier Dammond

```
clc; clear;

% ===== Define Average Link Lengths (mm) =====
L1 = 45; % Proximal phalanx: These are the average lengths for a nor
%human hand and not our hands.
L2 = 25; % Middle phalanx
L3 = 20; % Distal phalanx
L4 = 15; % Extra flexion for ring/pinky

% ===== Target Fingertip Positions (mm) =====
fingers = {'Thumb','Index','Middle','Ring','Pinky'};
targets = [ 60 20;
            70 10;
            80 15;
            75  5;
            65  0];

% ===== Results =====
angle_results = cell(length(fingers), 5);

for i = 1:length(fingers)
    finger = fingers{i};
    x = targets(i, 1);
    y = targets(i, 2);

    % Combine L2 + L3 for effective 2R solution
    Leff1 = L1;
    Leff2 = L2 + L3;

    % Inverse Kinematics
    r2 = x^2 + y^2;
    cos_theta2 = (r2 - Leff1^2 - Leff2^2) / (2 * Leff1 * Leff2);
    cos_theta2 = max(min(cos_theta2, 1), -1); % Clamp to [-1, 1]
    theta2 = acos(cos_theta2); % In radians

    k1 = Leff1 + Leff2 * cos(theta2);
    k2 = Leff2 * sin(theta2);
    theta1 = atan2(y, x) - atan2(k2, k1); % In radians

    % DIP joint estimate (proportional to PIP)
    theta3 = theta2 * (20/45); % Simplified approximation

    % Convert to degrees
    theta1_deg = rad2deg(theta1);
    theta2_deg = rad2deg(theta2);
    theta3_deg = rad2deg(theta3);
end
```

```

% For ring and pinky, add a 4th joint estimate
if strcmp(finger, 'Ring') || strcmp(finger, 'Pinky')
    theta4_deg = 20; % Fixed curl or tendon pull
else
    theta4_deg = NaN;
end

% Store results
angle_results{i,1} = finger;
angle_results{i,2} = round(theta1_deg, 2);
angle_results{i,3} = round(theta2_deg, 2);
angle_results{i,4} = round(theta3_deg, 2);
angle_results{i,5} = round(theta4_deg, 2);
end

% ===== Display Table =====
T = cell2table(angle_results, 'VariableNames', {'Finger', '01_deg', '02_deg', '03_deg', '04_deg'});
disp(T);

```

Finger	01_deg	02_deg	03_deg	04_deg
{ 'Thumb' }	-26.92	90.71	40.31	NaN
{ 'Index' }	-30.09	76.43	33.97	NaN
{ 'Middle' }	-14.64	50.52	22.45	NaN
{ 'Ring' }	-29.55	66.73	29.66	20
{ 'Pinky' }	-43.76	87.52	38.9	20